THE QUOTIENT MAP OF AN ETALE EQUIVALENCE RELATION IS ETALE AND SURJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays a three-term chain of equalities; it is part of the counted proof span and is preserved verbatim.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Description of morphisms into a quotient by an fppf covering. used at proof lines 1659-1667. Turns a morphism T -> U/R into the data (T_i, a_i).
- Surjectivity and etaleness are local on the base for the fpqc topology. Remark remark-list-properties-fpqc-local-base, used at proof lines 1676-1679. Descends the property from the covering to T.

Source reference. Original remark 02WH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3804–3804. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let U be a scheme over S. Let $j = (s,t) : R \to U \times_S U$ be an étale equivalence relation on U over S. If the quotient U/R is an algebraic space, then $U \to U/R$ is étale and surjective. Hence $(U, R, U \to U/R)$ is a presentation of the algebraic space U/R.*

Proof. Denote $c : U \to U/R$ the morphism in question. Let T be a scheme and let $a : T \to U/R$ be a morphism. We have to show that the morphism (of schemes) $\pi : T \times_{a,U/R,c} U \to T$ is étale and surjective. The morphism a corresponds to an fppf covering $\{\varphi_i : T_i \to T\}$ and morphisms $a_i : T_i \to U$ such that $a_i \times a_{i'} : T_i \times_T T_{i'} \to U \times_S U$ factors through R, and such that $c \circ a_i = a \circ \varphi_i$. Hence

$$T_i \times_{\varphi_i,T} T \times_{a,U/R,c} U = T_i \times_{c \circ a_i,U/R,c} U = T_i \times_{a_i,U} U \times_{c,U/R,c} U = T_i \times_{a_i,U,t} R.$$

Since t is étale and surjective we conclude that the base change of π to T_i is surjective and étale. Since the property of being surjective and étale is local on the base in the fpqc topology (see Remark 02WH) we win. □

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.